

■ Stock Finder

by Interest

Discover publicly traded companies based on what you care about — powered by AI and real market data.

AI Engine	Groq · Llama 3.3 70B	Data Source	Yahoo Finance (yfinance)
News Search	Tavily real-time API	Charts	Plotly interactive
Framework	Streamlit	Design	Glass & Glow dark UI

This document explains what the app does, how each feature works, what data it uses, and how the AI analysis is generated.

1. The Problem This Solves

Most people want to invest but feel shut out of the process. The standard advice — "do your research" — assumes you already know which companies to research and which metrics to look at. That's a high barrier.

This tool flips the starting point: instead of starting from a ticker symbol, you start from something you already know and care about — gaming, electric vehicles, space, AI chips. The app finds the relevant publicly traded companies and immediately gives you the data and analysis you need to evaluate them.

2. How It Works — End to End

Step	What happens	Technology
1	User types an interest topic	Streamlit text input
2	Topic validated (is it real?)	Groq Llama 3.3 70B
3	6 companies + reasons generated	Groq Llama 3.3 70B
4	Stock data fetched for each	yfinance (Yahoo Finance)
5	Cards displayed with price, change, cap, reason	Streamlit + custom HTML/CSS
6	User selects a stock	Streamlit session state
7	12 financial metrics displayed	yfinance .info dict
8	Interactive chart rendered	Plotly
9	Latest news searched	Tavily real-time API
10	AI analyst report generated	Groq Llama 3.3 70B

3. Features in Detail

3.1 Interest-Based Discovery

The user types a plain-English interest — anything from 'coffee' to 'quantum computing'. A validation step uses Llama 3.3 70B to confirm the topic is real and relates to publicly traded companies before spending API calls on company discovery.

- 6 companies returned per search, each with a one-line reason
- 'Show More' fetches 6 new companies, never repeating already-shown tickers
- Example chips (Gaming, EVs, AI & Chips, Space) for first-time users

3.2 Stock Cards

Each card shows enough to make a quick judgment without clicking through:

Field	Source	Why it matters
Company name + ticker	yfinance longName	Human-readable identity
Current price	yfinance currentPrice	Baseline for all relative metrics
Intraday change %	yfinance previousClose	What's happening right now
Market cap + size badge	yfinance marketCap	Risk profile (micro vs large cap)
AI reason	Groq LLM output	Why this stock matched your interest

3.3 Theme Portfolio View

A toggle above the grid switches to a normalised performance chart showing all 6 stocks rebased to 100 at the same start date. This lets the user compare which companies in their interest area have outperformed or underperformed over the past year.

- Best performer, laggard, and average return shown as summary stats
- Reference line at 100 shows the starting point
- Uses existing 1Y data — no extra API calls

3.4 12-Metric Financial Panel

The stock detail view shows three rows of four metrics each, all sourced from yfinance:

Metric	Meaning
Price + intraday Δ	Current price vs previous close
Market Cap	Total company value at current price
Trailing P/E	Price $\div$ last 12 months earnings — current valuation
Forward P/E	Price $\div$ next 12 months estimated earnings — growth expectations
52W High / Low	The trading range over the past year — where price sits now
Analyst Target	Wall Street consensus price target
Analyst Rating	Strong Buy / Buy / Hold / Sell consensus
Earnings Date	Next quarterly results — often the biggest near-term catalyst
Dividend Yield	Annual dividend as % of price — income vs growth signal

Beta	Volatility relative to S&P 500 (1.0 = same, 2.0 = twice as volatile)
Short Interest	% of float sold short — high = bearish sentiment or squeeze risk

3.5 Interactive Price Chart

Plotly renders an interactive dark-themed chart with seven time ranges (1D to All-time). The active range button is highlighted in blue. Each chart includes:

- Price line with colour-coded fill — green when up, red when down
- Volume bars on a secondary y-axis (lower panel)
- 50-day moving average overlay (orange dotted line) for ranges ≥ 1 month
- Unified hover tooltip showing price and volume at the same timestamp

3.6 AI Analysis (Real-Time)

When the user opens a stock's detail view, the app runs a two-step AI pipeline:

- **Step 1 — News retrieval:** Tavily searches for the 5 most recent news articles about the company, returning title, content snippet, source, and publication date.
- **Step 2 — Analyst report:** Groq (Llama 3.3 70B) receives the news text plus a financial snapshot (price, P/E, 52W range, analyst target, earnings date, beta) and generates a structured four-section analysis.

Section	What it covers
Upcoming Catalysts	2-3 near-term events (earnings, product launches, regulatory decisions) with dates and why they matter
Historical Comparison	1-2 similar past events for this company or sector with specific dates and % price moves
Price Impact Assessment	Balanced upside/downside scenarios anchored to the 52W range and analyst target
Key Risks	2-3 specific risks referencing beta (volatility), valuation, and competitive threats

The analysis is cached in session state — it only runs once per stock per session, so switching time ranges or revisiting the stock doesn't re-trigger the API calls.

4. Tech Stack

Library	Version	Role
<code>streamlit</code>	latest	Web framework, session state, UI components
<code>yfinance</code>	latest	Historical prices, volume, 50+ financial metrics via Yahoo Finance
<code>plotly</code>	latest	Interactive price + volume charts
<code>groq</code>	latest	LLM client — Llama 3.3 70B for company matching and analysis
<code>tavily-python</code>	latest	Real-time news search API
<code>python-dotenv</code>	latest	Load API keys from .env file

5. Setup & Running Locally

Step	Command / Action
1. Clone	<code>git clone <repo-url> && cd stock-interest-finder</code>
2. Install	<code>pip install -r requirements.txt</code>
3. Create .env	GROQ_KEY=your_key TAVILY_KEY=your_key
4. Run	<code>streamlit run app.py</code>
5. Open	<code>http://localhost:8501</code> (auto-opened by Streamlit)

API keys required: Groq (free tier available at console.groq.com) and Tavily (free tier available at tavily.com). No paid subscriptions needed to run the app.

6. UI Design System

The interface uses a Glass & Glow aesthetic — dark backgrounds with subtle glassmorphism cards, ambient radial glow lighting, and a blue/purple accent palette. The Inter typeface is loaded from Google Fonts for all text.

Element	Value	Purpose
Background	#060912	Deep dark — reduces eye strain for data-heavy screens
Ambient glow	Radial gradient	Blue at top-left, purple at bottom-right — depth
Card background	rgba(255,255,255,0.035)	Glassmorphism — semi-transparent with blur
Card border	rgba(255,255,255,0.07)	Subtle separation without harsh lines
Primary accent	#3b82f6 (blue)	CTAs, active states, links, ticker pills
Up / positive	#10b981 (green)	Price increases, positive metrics
Down / negative	#f43f5e (red/rose)	Price decreases, negative metrics
MA overlay	#f97316 (orange)	50-day moving average — visually distinct from price
Text primary	#f8fafc	Headings, prices, metric values
Text secondary	#94a3b8	Body copy, descriptions
Text dim	#475569	Labels, metadata, captions

7. Disclaimer

This tool is for research and educational purposes only. It does not constitute investment advice. The AI-generated analysis may be inaccurate, incomplete, or outdated. Always conduct your own due diligence and consult a qualified financial adviser before making any investment decisions.

Stock prices and financial data are sourced from Yahoo Finance via yfinance and may be delayed. AI analysis is generated by Llama 3.3 70B and is not verified by any financial professional.